

Daily Price & Inflation Tracking System for Cambodia

A simple, automated system that checks daily retail prices across Cambodia to provide real-time inflation updates without waiting 1 to 2 months for official survey reports.

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Why Are We Doing This?

The main problem with waiting for traditional monthly inflation reports

The Long Waiting Time

- **1 to 2 Months Delay:** Traditional inflation reports are only released once a month and usually take 30 to 60 days to publish after data collection.
- **Missing Daily Price Changes:** When prices of gasoline, rice, or food spike today, government leaders do not see the official numbers until weeks later.
- **Delayed Decision Making:** Long waiting periods make it difficult to take fast, timely action to help vulnerable families when prices rise suddenly.

Cambodia's Everyday Reality

- **High Spending on Essentials:** Over 70% of an average Cambodian family's daily budget is spent on food and transportation. These prices change frequently.
- **Two Currencies (USD & Khmer Riel):** Many stores display prices in US Dollars while others use Khmer Riel. Daily exchange rate changes directly affect prices.
- **The Need:** A modern, automated system that checks prices every single morning and calculates inflation in real time.

What Am I Doing?

Automatically gathering thousands of everyday prices across Cambodia every morning

20+

Online Sources Monitored

Supermarkets, pharmacies, fuel stations, bus travel, and housing

31,000+

Prices Collected Daily

Gathered automatically every morning before the workday starts

100%

Everyday Needs Covered

All official household spending categories included

What Kinds of Everyday Prices Are We Tracking?

- **Supermarkets & Groceries:** Major stores (AEON, DeliShop, Chip Mong, etc.) for rice, meat, cooking oil, eggs, and milk.
- **Health & Medicine:** Pharmacies for common medicines, vitamins, and family healthcare products.
- **Gasoline & Travel:** Daily retail fuel prices from the Ministry of Commerce and inter-city bus travel fares.
- **Official Exchange Rates:** The official daily USD/KHR rate directly from the Ministry of Economy and Finance.

How Do We Clean and Organize the Prices?

Turning raw online store information into clean, reliable numbers

1. Standardizing

Cleaning text and currency:

- Cleans up spelling errors and mixes of Khmer and English names.
- Converts all US Dollar prices into Khmer Riel using that day's official rate.
- Calculates the price per standard unit (e.g. price per 1 kg or price per 1 liter).

2. Smart Matching

Connecting identical products:

- Recognizes when two different stores sell the exact same item under slightly different names.
- Ensures small packages are never mixed up with larger family-size packages.
- Spots when package sizes shrink while the shelf price stays the same.

3. Official Categories

Sorting into national baskets:

- Sorts each product into official government categories (Food, Housing, Health, Transport, etc.).
- Categorizes thousands of products instantly using smart sorting rules.
- Ensures the data directly matches Cambodia's national survey standards.

Daily Inflation & Early Forecasting

Knowing price trends immediately instead of waiting for next month

Daily Inflation Updates

- **Calculated Every Morning:** The system automatically calculates price changes every day, showing whether living costs rose or fell.
- **Fair Calculation Rules:** Uses international rules so temporary weekend discounts and store sales do not create false inflation spikes.
- **Two Clear Views:**
 - *Overall Inflation:* Covers all everyday products families purchase.
 - *Core Inflation:* Excludes volatile fresh food and gasoline to show the underlying long-term trend.

Early Month-End Forecast

- **No Need to Wait 60 Days:** While the current month is still happening, the system forecasts where monthly inflation will land.
- **Watches Fast-Moving Goods:** It monitors rapid changes in food and fuel prices to predict the remaining days of the month.
- **Gets More Accurate Every Day:** As each day passes and more real data comes in, the forecast becomes narrower and more reliable.

Where Does the Project Stand Today?

The entire system is already built, fully operational, and running every morning

1. Automatic Daily Price Collection Over 20 sources and stores are collected automatically every morning at 2:00 AM.	✓ 100% Running Daily
2. Data Cleaning & Product Sorting Prices are cleaned, converted to Khmer Riel, and sorted into official categories automatically.	✓ 100% Operational
3. Daily Inflation & Early Forecasts Calculates daily price trends and produces early month-end inflation forecasts.	✓ 100% Implemented
4. Visual Charts & Dashboards Easy-to-read charts and reports are online for monitoring price trends and system health.	✓ 100% Live
5. Graduation Thesis Report All 5 report chapters are written; currently completing final formatting and review.	✎ Final Polish

Benefits for Cambodia & Next Steps

How this project helps leaders make better decisions and the path to graduation

How This Helps the Government

- **Early Warning System:** Leaders at the Ministry of Economy and Finance can spot sudden price spikes in food or fuel immediately.
- **Protects Families:** Quickly shows which essential goods are getting more expensive so supportive policies can be made earlier.
- **Digital Public Asset:** Demonstrates how modern digital technology can enhance national economic tracking in Cambodia.

Next Steps to Completion

- **Next Week:** Review presentation slides and results with ministry supervisors.
- **Late September:** Submit the finalized graduation report to the university committee.
- **September 30, 2026:** Final Thesis Defense at the Institute of Technology of Cambodia (ITC).
- **System Handover:** Deliver the automated system and user guides to the ministry team for continued daily operation.